

K_n Values for Equation (2)

Table A.1. Equipment Cost Data to Be Used with [Equation \(A.1\)](#)

Equipment Type	Equipment Description	K_1	K_2	K_3	Capacity, Units	Min Size	Max Size
Blenders	Kneader	5.0141	0.5867	0.3224	Volume, m ²	0.14	3
	Ribbon	4.1366	0.5072	0.0070	Volume, m ²	0.7	11
	Rotary	4.1366	0.5072	0.0070	Volume, m ²	0.7	11
Centrifuges	Auto batch separator	4.7681	0.9740	0.0240	Diameter, m	0.5	1.7
	Centrifugal separator	4.3612	0.8764	-0.0049	Diameter, m	0.5	1
	Oscillating screen	4.8600	0.3340	0.1063	Diameter, m	0.5	1.1
	Solid bowl w/o motor	4.9697	1.1689	0.0038	Diameter, m	0.3	2
Compressors	Centrifugal, axial, and reciprocating	2.2897	1.3604	-0.1027	Fluid power, kW	450	3000
	Rotary	5.0355	-1.8002	0.8253	Fluid power, kW	18	950
Conveyors	Apron	3.9255	0.5039	0.1506	Area, m ²	1.0	15
	Belt	4.0637	0.2584	0.1550	Area, m ²	0.5	325
	Pneumatic	4.6616	0.3205	0.0638	Area, m ²	0.75	65
	Screw	3.6062	0.2659	0.1982	Area, m ²	0.5	30
Crystallizers	Batch	4.5097	0.1731	0.1344	Volume, m ³	1.5	30
Drives	Gas turbine	-21.7702	13.2175	-1.5279	Shaft power, kW	7500	23,000
	Intern comb. engine	2.7635	0.8574	-0.0098			
	Steam turbine	2.6259	1.4398	-0.1776			
	Electric—explosion-proof	2.4604	1.4191	-0.1798			
	Electric—totally enclosed	1.9560	1.7142	-0.2282			
	Electric—open/drip-proof	2.9508	1.0688	-0.1315			

$$\log_{10} C_p^0 = K_1 + K_2 \log_{10}(S) + K_3 [\log_{10}(S)]^2$$

Notice the size range for each equipment!

Heat exchangers	Scraped wall	3.7803	0.8569	0.0349	Area, m ²	2	20
	Teflon tube	3.8062	0.8924	-0.1671	Area, m ²	1	10
	Bayonet	4.2768	-0.0495	0.1431	Area, m ²	10	1000
	Floating head	4.8306	-0.8509	0.3187	Area, m ²	10	1000
	Fixed tube	4.3247	-0.3030	0.1634	Area, m ²	10	1000
	U-tube	4.1884	-0.2503	0.1974	Area, m ²	10	1000
	Kettle reboiler	4.4646	-0.5277	0.3955	Area, m ²	10	100
	Double pipe	3.3444	0.2745	-0.0472	Area, m ²	1	10
	Multiple pipe	2.7652	0.7282	0.0783	Area, m ²	10	100
	Flat plate	4.6656	-0.1557	0.1547	Area, m ²	10	1000
	Spiral plate	4.6561	-0.2947	0.2207	Area, m ²	1	100
	Air cooler	4.0336	0.2341	0.0497	Area, m ²	10	10,000
	Spiral tube	3.9912	0.0668	0.2430	Area, m ²	1	100
	Diphenyl heater	2.2628	0.8581	0.0003	Duty, kW	650	10,750
Heaters	Molten salt heater	1.1979	1.4782	-0.0958	Duty, kW	650	10,750
	Hot water heater	2.0829	0.9074	-0.0243	Duty, kW	650	10,750
	Steam boiler	6.9617	-1.4800	0.3161	Duty, kW	1200	9400
Mixers	Impeller	3.8511	0.7009	-0.0003	Power, kW	5	150
	Propeller	4.3207	0.0359	0.1346	Power, kW	5	500
	Turbine	3.4092	0.4896	0.0030	Power, kW	5	150
Packing	Loose (for towers)	2.4493	0.9744	0.0055	Volume, m ³	0.03	628
Process vessels	Horizontal	3.5565	0.3776	0.0905	Volume, m ³	0.1	628
	Vertical	3.4974	0.4485	0.1074	Volume, m ³	0.3	520
Pumps	Reciprocating	3.8696	0.3161	0.1220	Shaft power, kW	0.1	200

Purchase Cost (C_p^0) Plots

Data also
presented as
plots in
**Figures A.1 to
A.17**

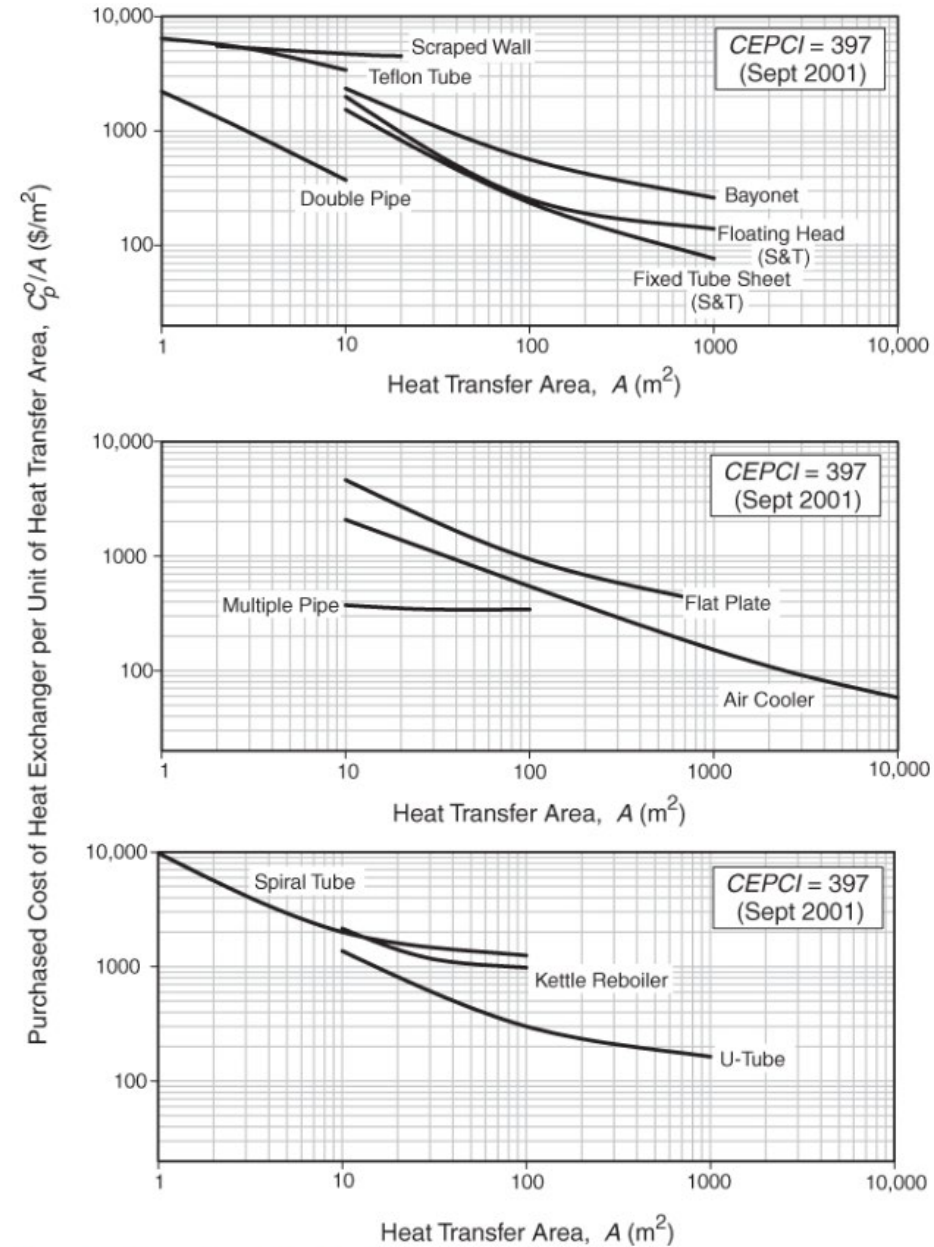


Figure A.5. Purchased Costs for Heat Exchangers

Pressure Factor Constants (C_n) for Equation (3)

Table A.2. Pressure Factors for Process Equipment (Correlated from Data in Guthrie [1, 2], and Ulrich [3])

Equipment Type	Equipment Description	C_1	C_2	C_3	Pressure Range (barg)
Compressors	Centrifugal, axial, rotary, and reciprocating	0	0	0	—
Drives	Gas turbine	0	0	0	—
	Intern. comb. engine	0	0	0	—
	Steam turbine	0	0	0	—
	Electric—explosion-proof	0	0	0	—
	Electric—totally enclosed	0	0	0	—
	Electric—open/drip-proof	0	0	0	—
Evaporators	Forced circulation (pumped), falling film, agitated film (scraped wall), short tube, and long tube	0	0	0	P<10
		0.1578	-0.2992	0.1413	10<P<150
Fans*	Centrifugal radial, and centrifugal backward curve	0	0	0	$\Delta P < 1 \text{ kPa}$
		0	0.20899	-0.0328	$1 < \Delta P < 16 \text{ kPa}$
	Axial vane and axial tube	0	0	0	$\Delta P < 1 \text{ kPa}$
		0	0.20899	-0.0328	$1 < \Delta P < 4 \text{ kPa}$
Furnaces	Reformer furnace	0	0	0	P<10
		0.1405	-0.2698	0.1293	10<P<200
	Pyrolysis furnace	0	0	0	P<10
		0.1017	-0.1957	0.09403	10<P<200
	Nonreactive fired heater	0	0	0	P<10
		0.1347	-0.2368	0.1021	10<P<200
Heat exchangers	Scraped wall	0	0	0	P<40
		0.6072	-0.9120	0.3327	40<P<100
		13.1467	-12.6574	3.0705	100<P<300
	Teflon tube	0	0	0	P<15
	Bayonet, fixed tube sheet, floating head, kettle reboiler, and U-tube (both shell and tube)	0	0	0	P<5
		0.03881	-0.11272	0.08183	5<P<140
	Bayonet, fixed tube sheet, floating head, kettle reboiler, and U-tube (tube only)	0	0	0	P<5
		-0.00164	-0.00627	0.0123	5<P<140
	Double pipe and multiple pipe	0	0	0	P<40

$$\log_{10} F_P = C_1 + C_2 \log_{10}(P) + C_3 [\log_{10}(P)]^2$$

Notice the pressure range over which the correlations are valid!
Extrapolation outside these pressures should be done with extreme caution.

Material Factors, F_M (ID No. and Plot)

Table A.3. Identification Numbers for Material Factors for Heat Exchangers, Process Vessels, and Pumps to Be Used with [Figure A.18](#)

Identification Number	Equipment Type	Equipment Description	Material of Construction
1	Heat exchanger	Double pipe, multiple pipe,	CS-shell/CS-tube
2		fixed tube sheet, floating head,	CS-shell/Cu-tube
3		U-tube, bayonet, kettle reboiler, scraped	Cu-shell/Cu-tube
4		wall, and spiral tube	CS-shell/SS-tube
5			SS-shell/SS-tube
6			CS-shell/Ni alloy tube
7			Ni alloy, shell/Ni alloy-tube
8			CS-shell/Ti-tube
9			Ti-shell/Ti-tube
10	Process vessels	Air cooler	CS tube
11		Air cooler	Al tube
12		Air cooler	SS tube
13		Flat plate and spiral plate	CS (in contact with fluid)
14		Flat plate and spiral plate	Cu (in contact with fluid)
15		Flat plate and spiral plate	SS (in contact with fluid)
16		Flat plate and spiral plate	Ni alloy (in contact with fluid)
17		Flat plate and spiral plate	Ti (in contact with fluid)
18		Horizontal, vertical (including towers)	CS
19		Horizontal, vertical (including towers)	SS clad
20		Horizontal, vertical (including towers)	SS
21		Horizontal, vertical (including towers)	Ni alloy clad
22		Horizontal, vertical (including towers)	Ni alloy
23		Horizontal, vertical (including towers)	Ti clad
24		Horizontal, vertical (including towers)	Ti
25	Pumps	Reciprocating	Cast iron
26		Reciprocating	Carbon steel
27		Reciprocating	Cu alloy

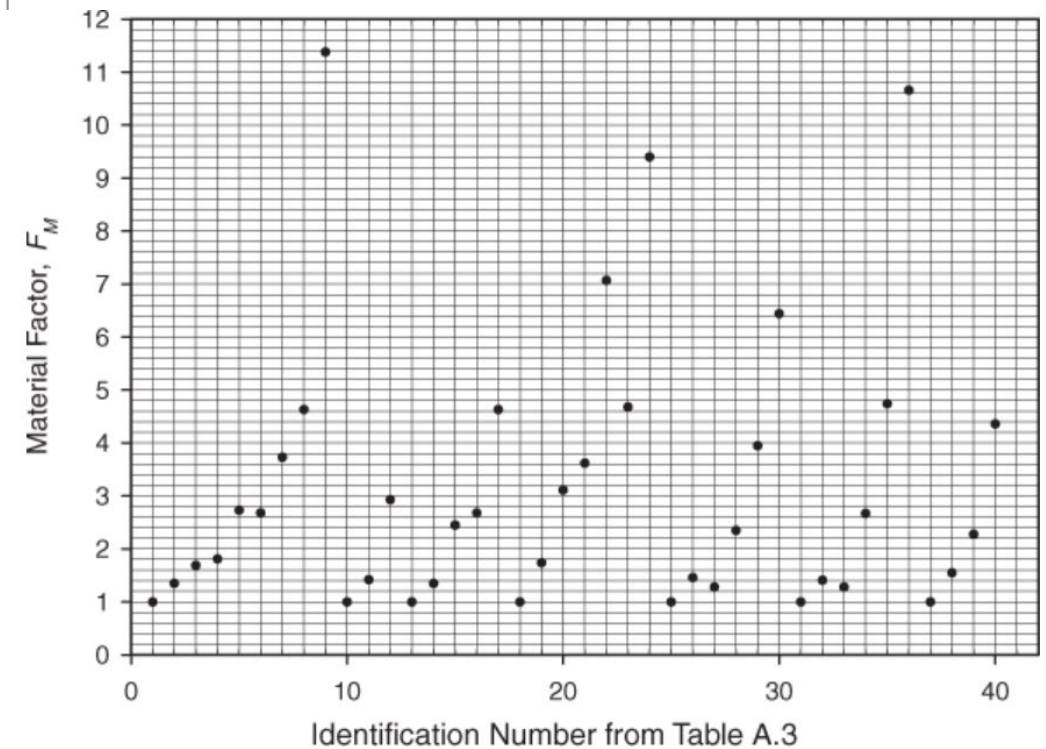


Figure A.18. Material Factors for Equipment in [Table A.3](#) (Averaged Data from References [\[1, 2, 3, 6, 7, and 8\]](#))

For different
equipment
with different
MOC

Constants for Equation (1)

Table A.4. Constants for Bare Module Factor to Be Used in [Equation \(A.4\)](#)
(Correlated from Data in Guthrie [\[1, 2\]](#) and Ulrich [\[3\]](#))

Equipment Type	Equipment Description	B_1	B_2
Heat exchangers	Double pipe, multiple pipe, scraped wall, and spiral tube	1.74	1.55
	Fixed tube sheet, floating head, U-tube, bayonet, kettle reboiler, and Teflon tube	1.63	1.66
	Air cooler, spiral plate, and flat plate	0.96	1.21
Process vessels	Horizontal	1.49	1.52
	Vertical (including towers)	2.25	1.82
Pumps	Reciprocating	1.89	1.35
	Positive displacement	1.89	1.35
	Centrifugal	1.89	1.35

$$C_{BM} = C_P^0 (B_1 + B_2 F_P F_M)$$

Bare Module Factors for Other Equipment

Table A.7. Bare Module Factors for Conveyors, Crystallizers, Dryers, Dust Collectors, Filters, Mixers, Reactors, and Screens

Equipment Type	Equipment Description	Bare Module Factor, F_{BM}
Blenders	Kneader	1.12*
	Ribbon	1.12*
	Rotary	1.12
Centrifuges	Auto batch separator	1.57*
	Centrifugal separator	1.57
	Oscillating screen	1.57*
	Solid bowl w/o motor	1.27
Conveyors	Apron	1.20
	Belt	1.25
	Pneumatic	1.25*
	Screw	1.10
Crystallizers	Batch	1.60
Dryers	Drum	1.60
	Rotary, gas fired	1.25
	Tray	1.25
Dust Collectors	Baghouse	2.86*
	Cyclone scrubbers	2.86*
	Electrostatic precipitator	2.86*
	Venturi scrubber	2.86*
Filters	Bent	1.65*
	Cartridge	1.65*
	Disc and drum	1.65*
	Gravity	1.65*
	Leaf	1.65
	Pan	1.65*
	...	...
	...	...

Bare module factors
for equipment added
in third edition of the
book (includes
reactors)

Bare Module Cost Equations for Other Equipment

Table A.5. Equations for Bare Module Cost for Equipment Not Covered by
Tables A.3 and A.4

Equipment Type	Equation for Bare Module Cost
Compressors and blowers without drives	$C_{BM} = C_p^o F_{BM}$
Drives for compressors and blowers	$C_{BM} = C_p^o F_{BM}$
Evaporators and vaporizers	$C_{BM} = C_p^o F_{BM} F_P$
Fans with electric drives	$C_{BM} = C_p^o F_{BM} F_P$
Fired heaters and furnaces	$C_{BM} = C_p^o F_{BM} F_P F_T$
F_T is the superheat correction factor for steam boilers ($F_T = 1$ for other heaters and furnaces) and is given by $F_T = 1 + 0.00184\Delta T - 0.00000335(\Delta T)^2$ where ΔT is the amount of superheat in °C.	
Power recovery equipment	$C_{BM} = C_p^o F_{BM}$
Sieve trays, valve trays, and demister pads	$C_{BM} = C_p^o N F_{BM} F_q$ where N is the number of trays and F_q is a quantity factor for trays only given by $\log_{10} F_q = 0.4771 + 0.08516 \log_{10} N - 0.3473 (\log_{10} N)^2 \text{ for } N < 20$ $F_q = 1 \text{ for } N \geq 20$
Tower packing	$C_{BM} = C_p^o F_{BM}$

Bare module factors for these equations given in Figure A.19 with ID No.'s in Table A.6